

SCHEDULE A – ATTACHMENT 1 – BUSINESS SPECIFICATION WORKSHEET

Instructions for Completing the Business Specifications Worksheet

Contractors must respond to each business Specification on how they will meet the requirements in the document provided. Contractor must not alter the document.

The Business Specifications Worksheet contains columns and is defined as follows:

Column A: Business Specification number.

Column B: Business Specification description.

Column C: Bidder must indicate how it will comply with the Business Specification. Contractor must enter "X" to one of the following:

- **Current Capability** – This capability is available in the proposed Solution with no additional configuration or cost
- **Not Available** – This capability is not currently available.

Column D: The Bidder must also fully disclose how they will meet the requirements in their proposal response. This column is for Bidder to describe how they will deliver the business Specification. Description must be no more than 250 words for each Business Specification.

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Business Specification Number	Business Specification	Current Capability	Not Available	Contractor to explain how they will deliver the Business Specification.
Base Level Criteria				
1.0	Unified self-service experience: All components of the offered services exist within the same portal and service catalog, in a fully integrated fashion — not merely as part of an “umbrella” portal that contains subportals for the offerings. The (Command Line Interface (CLI) and API (Application Programming Interface) will be consistent across these components. Any Integrated Development Environment (IDE) integration should be similarly	■		

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	consistent. They look and feel like a seamless whole. This provides a better user experience than nonintegrated offerings. It is permissible to have an additional, specialized portal experience for a capability, with greater depth of functionality than is exposed through the integrated offering's portal.		
2.0	Unified identity and access management: The same identity is used to access all components of the offered services, with unified authentication and authorization (including Role Based Access Control (RBAC)) for the portal, CLI and API. The use of a single IAM system prevents the need for either the Contractor or the State to federate identity across components of the offered services. This ensures that identity can be managed across the solution as a unified whole, thus simplifying user management as well as security.	■	
3.0	Single integrated low-latency network context: The cloud service components will exist within the same network, with low local-area network (LAN) latency. The components will be able to communicate within the boundaries of a secure, microsegmented, State-	■	

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	controlled, software-defined virtual network. This will allow the State to design unified network topologies and apply consistent network security across the components. However, services do not need to support private networks unless otherwise noted.		
4.0	Unified security: The State will be able to apply security controls across all components of the Cloud Services. The security context will be unified so that there is visibility and control across the entire environment.	■	
5.0	Enterprise-class support: The Contractor will offer a support option that includes 24/7 coverage, live support via phone and instant messaging, incident response times of 15 minutes or less, and a dedicated technical account manager. The State will be able to submit, review, and update billing and technical support tickets through the portal.	■	

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			[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
6.0	RBAC: The Contractor will have an Identity Access Management (IAM) service. The State will be able to have multiple users and API keys. The Contractor will have a self-service capability that allows the State to assign granular permissions to these identities.	■	[REDACTED] [REDACTED]
Resilience			
7.0	Data centers: The Contractor will must have three regions within the United States, separated by a minimum of 200 km (125 miles apart) and on different power grids.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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10.0	Continuously available critical services: The control plane (the software that controls the services and provides internal and external APIs), IAM service and KMS are critical services. They will be architected for resilience and continuous availability, and distributed across regions. No customer may have a critical service dependency on a single logical or physical data center, even if the service can fail over to another data center. The Contractor will be able to conduct maintenance activities without making any critical service temporarily unavailable to any customer. This includes continuous availability for all self-service interfaces.	■	
11.0	Monitoring service: The Contractor will offer a serverless monitoring service that, at minimum, contains the following capabilities: • Collectors: There will be a collector capable of ingesting metrics, events or other monitoring data from other services. Ingestion is required in	■	

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15.0	Initial administrative access credentials: The default administrative access credentials for every newly deployed compute instance will be automatically generated and able to be reset or chosen at the point of provisioning by the State. Automatically generated credentials will be retrievable in a secure fashion appropriate to the self-service provisioning mechanism. Email is unacceptable.	■	[REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
16.0	Trusted boot: The Contractor will ensure firmware security on its servers. The Contractor will use a hardware-based root of trust (for example, a security co-processor or trusted platform module [TPM]), secure boot (per the Unified Extensible Firmware Interface [UEFI] specification, or another means of preventing firmware updates from the main system processors), and measured boot. To accomplish the measured boot, the Contractor will securely store checksums for each system — thus enabling attestation to whether the system has changed since the last boot — and verify checksums prior to each boot.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED]

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			[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
17.0	Identity federation and SSO: The Contractor will support federated identity between The State's directory and its native IAM. The Contractor will allow single sign-on (SSO) login access to the portal, CLI and API through Security Assertion Markup Language (SAML) assertions or via a third-party SSO service that supports OpenID Connect.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
18.0	MFA administrative access control: There will be the option to secure both portal and API access via multifactor authentication (MFA), via the Contractor's native IAM. This may use either hardware- or software-based secure tokens, and HOTP or TOTP (hashed message authentication code [HMAC]-based or time-based one-time passwords). At least one Multi-Factor Authentication method available and configurable as required will support compliance with all NIST 800-63 AAL3 requirements. To secure API access via MFA, it is permissible for the user to use an MFA-enabled authentication method to retrieve a secure token that can then be used for API calls.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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19.0	Granular RBAC policies: All services will support RBAC via the Contractor's IAM service. Policies will be definable at the following levels of granularity: Platform-level: A role can have a permission-set that applies broadly across the whole platform. For instance, users might be either full users ("read/write" everything) or read-only users (typically used for management and billing admins). Service-level: A role can have a permission set that applies broadly across one service within the platform. For instance, the compute, storage and database services might each offer administrator, developer and billing-admin permission sets. API-level: A role can be given access to a specific API call (or equivalently narrow portal functionality). This may be at an even more granular level — for instance, particular restrictions on use of that API call, such as the ability to call it only with certain parameters.	■	
20.0	RBAC policies for resource groups: The State will be able to create an RBAC policy that applies to a resource group. Groups consist of multiple elements that have been grouped together and collectively given an associated keyword, like "mobile." Groups are typically used to associate elements with a particular business project (such as grouping all infrastructure associated	■	

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Operations and Governance			
22.0	Basic cost management: The Contractor will offer, at no cost, basic cloud service expense management. This will include the following capabilities, with billing data visibly updated at least every 24 hours: Tagged bills: The State will be able to access billing data for the previous and current month at any time. Exports of billing data will be in a programmatically readable format such as CSV, and include the State's resource groups or tags for each line item. Budgets: The State will be able to set a per-partition monthly budget, and generate an alert when the budget is projected to be exceeded as well as when the budget is actually exceeded.	■	[REDACTED]
23.0	Cost forecasting: The State will be able to forecast what they are likely to spend. The Contractor will offer a web-based calculator or downloadable Excel-compatible spreadsheet that predicts the monthly cost of a solution based on a State-defined model. This model will be able to include, at minimum, core infrastructure assets, KMS and monitoring. Finally, the Contractor will provide a pricing API, so the State can programmatically obtain current costs for all services.	■	[REDACTED]

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Vendor Management and Audit			
24.0	Service health dashboard: The Contractor will offer a public dashboard that displays the health of each of its services, on a regional basis, updated at least every five minutes. Current health will also be programmatically quarriable. During an incident, the dashboard will also display information about the incident. The Contractor will make a root cause analysis (RCA) available for all major outages.	■	[REDACTED]
25.0	Event management: The Contractor will offer at least one support plan that includes event management, for use with important events such as a planned application release or high-volume usage period (e.g., the holiday shopping period). This will include pre-event planning, guaranteed fast support response during the event, and post-event review. In addition, the Contractor will offer a support plan in which the State and Contractor agree to and practice a communication and response plan for crisis scenarios. This plan will include time-critical actions to be performed by the Contractor in the case of State unavailability.	■	[REDACTED]
26.0	Support ticket API: The Contractor will have a State-facing API for the case management system that is used for billing and technical support requests. At minimum, there will be APIs to open a ticket, update a ticket, close	■	[REDACTED]

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	a ticket, view (obtain fields of) a ticket, and list all open tickets.		[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
27.0	Release notes: The State will be able to track new services and features. The Contractor will publicly publish online release notes that list all new and updated capabilities. Each item will be dated and tagged with the affected services. The State will be able to easily see a chronological list of changes to each service. Release notes will be available for a minimum of three years.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
28.0	Decommissioning Account: The Contractor will allow the State to close an account in its entirety. Doing so will immediately stop all cost-accruing services and stop billing, with no need for the State to stop anything below the account level. The State will have the ability to require interaction with support or account management to initiate or confirm account closure.	■	[REDACTED] [REDACTED]
29.0	Decommissioning Order: The Contractor will allow the State to close an order in part or in its entirety. Doing so will immediately stop all cost-accruing services and stop billing.	■	[REDACTED] [REDACTED]

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30.0	Service change notification: The Contractor will implement a minimum of one year of notice for any “breaking changes.” This will include service discontinuations, changes that break backward compatibility or remove features in active use, and API changes that require the State to update its code. The terms of service will guarantee this notification.	■	[REDACTED]
31.0	Published compliance assistance: The Contractor will offer published guidance that supports the State in meeting common compliance needs. The Contractor will publicly publish a consolidated list of all compliance programs that its services can support, along with detailed supporting documentation for obtaining compliance in each program.	■	[REDACTED]
Technical Requirements			
1.0	All State Data for IT Environment Services including backups and Disaster Recovery will reside within the United States of America.	■	[REDACTED]
2.0	1. If the State requires FTI (Federal Tax Information) data to be received, transmitted, stored, or processed by the CSP, the Cloud Service		[REDACTED]

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	Provider will sign form Pub 1075, Exhibit 7 with the State. 2. If the State requires STI (State Tax Information) data to be received, transmitted, stored, or processed by the CSP, the Cloud Service Provider will sign Michigan form "SAFEGUARD REQUIREMENTS OF CONFIDENTIAL TAX DATA" with the State. See "SCHEDULE E, Attachment 1 – Tax Regulation"		
3.0	If the State requires CJIS (Criminal Justice Information System) data to be received, transmitted, stored, or processed by the CSP, the Cloud Service Provider will sign a Security Addendum with Michigan State Police (MSP). See "SCHEDULE E, Attachment 2 - FEDERAL BUREAU OF INVESTIGATION CRIMINAL JUSTICE INFORMATION SERVICES SECURITY ADDENDUM"		[REDACTED] [REDACTED] [REDACTED]
4.0	If the State requires Centers for Medicare & Medicaid Services (CMS) data to be received, transmitted, stored, or processed by the CSP, the Cloud Service Provider will sign a Security Agreement with the State per CMS MARS-E requirements.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
5.0	If the State desires to host data within the CSP which is subject to other similar security standards, the Cloud	■	[REDACTED] [REDACTED] [REDACTED]

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	Service Provider will sign a security agreement according to such standards.		
6.0	Contractor will provide a link to rate calculator that will show all services to the State and their current rate and will include any fixed or periodic subscription costs, and for consumption-based costs, the billing unit, the cost per billing unit, any information on minimum orderable quantities, minimum billable time segments, or tiered pricing, along with the following information for each service:	■	

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	Service type	Information required		
	Compute	Cloud region; processor family and type; processor speed; processor and core counts; total or per core memory quantity in GB; supported and excluded operating system families and versions		
	Storage	Cloud region; storage type; connection type; read and write speeds; high availability/redundancy service level; data backup and recovery service level, ingress, egress, and bulk data migration support		
	Network	Sustained and, if applicable, burst circuit speeds for both upload and download; physical and virtual dedicated connections, included and excluded network features such as routing, VLANs, firewalls, inspection and associated costs		
	Relational database services	Cloud region; database family and type; processor or core count; memory quantity in GB; high availability/redundancy service level; data backup and recovery service level		
	Containers	Cloud region; clusters, Instance Series; supported and excluded operating system families and version; managed OS Disks in GB;		
	Data warehouse	Cloud region; database family and type; processor or core count; memory quantity in GB; high availability/redundancy service		

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	service s level; data backup and recovery service level Functions as a service Cloud region; memory quantity in GB; connection speed; high availability/redundancy service level; included and excluded web service features such as libraries and frameworks, WAF, SSL, traffic inspection, and associated costs <i>Please note: It is the responsibility of the Contractor to continually provide accurate, up-to-date, and understandable service and rate information. If the State cannot understand the costs for offered Cloud Services, accurately and fairly compare them to those of other Contractors, and avoid unanticipated limitations or unexpected costs, we are under no obligation to accept them or select them when deploying solutions.</i> Rate sheets will be provided in a format that facilitates easy ingestion into the State's rate engine.		
7.0	Contractor provides a reporting console which will allow the State to view current and projected usage/ consumption.	■	

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8.0	Direct Cloud Service Providers contracted services will be listed as ‘FedRAMP Authorized’ in the General Services Administration (GSA) FedRAMP Marketplace at Impact Level ‘Low’, ‘Moderate’, or ‘High’ and a Service Model that includes ‘IaaS’ and ‘PaaS’. The authorization will be continuously maintained by the Contractor throughout participation in the program. Changes to FedRAMP status will be communicated to DTMB by Contractors within seven days, and will result in a reevaluation by the State of their qualifying status. Until their compliance status is validated by the State, no new services will be initiated with the Contractor.	■	[REDACTED]
9.0	The Contractor will furnish, in a format that facilitates easy ingestion into the Common Controls Framework (CCF), the set of controls from the catalog in NIST SP 800-53 r4 and r5 within one year of release, Security and Privacy Controls for Federal Information Systems and Organizations, that have been attested by a third-party assessment organization (3PAO) as either provided or inheritable from the provided Cloud Services. Changes to this control set will be made available via self-service portal so that the CCF can be updated to reflect the change and the State can communicate them to its customers at will. Compliance controls required outside the FedRAMP framework (such as from Pub 1075, MARS-E, etc) will also be validated by a qualified independent	■	[REDACTED]

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	third-party audit organization and provide in a similar format to 800-53 controls.		██ ██ ██ ██ ██ ██ ██ ██
10.0	Unified billing and support: The Cloud Services capabilities are part of a single commercial offering construct, under a single brand. The State will be able to buy all of the procured cloud service components on a single contract, with the exception of specific Software as a Service, offerings	■	██ ██ ██ ██ ██ ██
Resilience			
11.0	Metrics for core services: The Contractor's core services will have the following built-in monitoring metrics: • Compute metrics: Health (up/down) status, CPU and memory utilization, disk and network I/O. • Block storage service metrics: Bytes read and written, and I/O operations. • Load balancer (LB) service metrics: Request count, request latency, and a measure of successful and failed back-end requests. The latter should preferably offer a count of response codes received from the back-end by category (2xx, 3xx, 4xx, 5xx). • Relational DBaaS metrics: Database connections and latency. If the database service is not	■	██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██ ██

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	seamlessly scalable, the monitoring tool will also report I/O operations, CPU utilization and memory utilization.		
12.0	Compute instance anti-affinity: Anti-affinity rules help the State achieve greater application availability by reducing the possibility that a single physical failure event will impact multiple instances. The Contractor's multitenant VM compute instance offering will support: • Automatic anti-affinity: When the State provisions a new VM, the default placement algorithm will automatically attempt to place the VM on a physical host that isn't used by any of the State's other VMs. With this approach, the State's VMs are distributed among as many different physical hosts as possible (but are not guaranteed to be on different physical hosts). • Explicit host anti-affinity: The State will be able to explicitly set anti-affinity rules for individual VMs that need to be placed on different physical hosts from one another. The physical hosts will be within the same logical data center. • Explicit power anti-affinity: The State will be able to explicitly specify that compute instance "A" be in a different section of the same logical data center than instance "B," such that the two instances do not share fully identical power dependencies. The Contractor may offer this for groups of instances rather than individual instances.	■	
13.0	VM-preserving host maintenance: The Contractor will be able to perform maintenance on a compute host with	■	

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	minimal disruption to running VMs on that host. Such maintenance includes performing a kernel upgrade or a hypervisor upgrade. Minimizing VM disruption may be accomplished by a variety of means, such as live migration to another host or memory-preserving maintenance.		
14.0	VM-preserving data center maintenance: If the Contractor needs to perform maintenance that impacts the physical compute host, it will do so without disruptively impacting running VMs on that host. Such maintenance includes replacing and upgrading hardware, and performing power maintenance that eliminates power to the host. Live migration would be the standard way to avoid VM disruption. The State does not need to have any control over this capability.	■	
15.0	VM host failure recovery: The VM service will be architected to automatically restart VMs on a healthy host if the original physical host fails. This capability is often referred to as "VM restart." The service may automatically attempt to reboot the original physical host and restart VMs on it. However, if that fails, the service will be able to automatically restart the VMs on a healthy host. The failure detection will occur within one minute, and if the host cannot be immediately recovered, the process of automatically restarting the VM on another host will begin within five minutes.	■	

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Security			
16.0	Key management service: The Contractor will offer a KMS that allows the State to create, control, and audit cryptographic key usage. It will facilitate life cycle management of keys (key expiration and rotation). The service will enable the State to comply with the National Institute of Standards and Technology (NIST) Federal Information Processing Standard (FIPS) 140-2 and applicable NIST key management publications. The cryptographic component will be FIPS 140-2 validated (independently audited), at a minimum of Level 1. The KMS will support BYOK and may support HYOK; the State will be able to import their own master keys into the KMS, including keys they have generated on-premises.	■	
17.0	Certificate management service: The Contractor will offer a service that manages obtaining, renewing and deploying FIPS compliant SSL/TLS certificates. This will support certificates from a certificate authority (CA) and not just self-signed certificates. This will be integrated with all load-balancing services that terminate SSL/TLS if required when configured by the State, as well as the CDN (if it supports SSL/TLS).	■	
18.0	Stateful network firewall: The State will be able to filter ingress and egress traffic — including both north-south and east-west traffic — via firewall rules. The State will be able to create multiple rules associated with each subnet and vNIC. The State will also be able to group compute instances and associate multiple rules with the group rules Traffic filtering tools will be stateful. All rule updates will apply immediately, without reprovisioning or restarting any services. It is permissible to provide these capabilities as part of different services.	■	

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19.0	Network forensics: The Contractor will enable the State to log metadata about network traffic that is permitted and denied by firewall rules and other network policies. The data will be available at an interval of 10 minutes or less, and the State will be able to export the data for external analysis. The State will be able to generate alerts based on logged events.	■	[REDACTED]
Operations and Governance			
20.0	Completeness of self-service interfaces: The Contractor will support all capabilities in the API, CLI and portal, to the extent that each capability makes sense in all three modalities. The Contractor will publicly publish a policy that indicates that new capabilities that are initially released only as APIs will be represented in the portal and CLI within 180 days of GA release. In addition, the Contractor will support SDKs that wrap its APIs in at least three different programming languages, with documentation and sample code. The Contractor will offer free plug-ins/extensions for integrated development environments (IDEs); these will, at minimum, support deploying applications into VMs. The CLI will be available for Linux, macOS, and Windows.	■	[REDACTED]

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			■ [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED]
21.0	Metadata tagging of resources: The State will be able to set metadata tags on all core infrastructure assets. A tag consists of a State-defined key and a value. The State will be able to attach at least seven tags per asset, with no limit on the total number of tags per partition. All tags will be searchable and filterable within all service interfaces. Tags will be visible and exportable within billing interfaces.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED]
22.0	Centralized policy management: The Contractor will offer a centralized capability for managing multiple partitions. The State will be able to organize partitions into a multilevel branching hierarchy at least three levels deep below the root of the master account, apply policies at the upper levels, and have those policies cascade down automatically. The State will be able to apply policies that restrict a partition's access to particular services. This will include the ability to enforce configuration rules for cloud resources. The Contractor will allow policies and standards to be set and enforced	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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	on a per-partition basis, including per-partitions exceptions to policies inherited from higher up in the hierarchy. This may be implemented as a centralized service that serves as an umbrella over multiple services that handle different types of policies, or incorporated across the Contractor's IAM capabilities.		
23.0	Distributed logging service: The Contractor will offer a highly scalable logging service that aggregates the State's application and system logs into a consistently time-ordered stream. The State will be able to centrally view, search, filter and archive the logs, as well as generate alerts based on pattern-matching against log entries. Log ingestion may use mechanisms like Fluentd, a protocol such as syslog, or a compute instance agent. The State will be able to stream log events in real time to other data-ingestion services or State industry standard Security tools for processing. The Contractor will also	■	

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	provide other security information and event management (SIEM) capabilities including support for State use of Contractor tools. It will integrate with other SIEM products.		[REDACTED]
24.0	Audit logs: The Contractor will offer a central service that aggregates logs of portal, CLI and API actions that result in provisioning or deprovisioning resources, access control changes, or changes to service configurations. At minimum, this will support all compute services, all storage services and other services that store data (such as database services), all networking, and all security services. The State will also be able to log all IAM actions in detail, and all logged actions will be associated with a responsible identity. The State will be able to export these logs for retention.	[REDACTED]	[REDACTED]

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25.0	Provisioning templates: The Contractor will offer an orchestration template service that provisions multiple resources and services, including a combination of compute instances, block storage volumes, networking elements, software infrastructure services, monitoring configurations and security configurations. The State will be able to define dependencies including authorization (so provisioning occurs in the correct order), provide parameters at provisioning time and use conditionals within the template. The Contractor will enable provisioning a template across multiple logical data centers and regions. The service will offer a template catalog, and access will be governed via RBAC.	■	
26.0	Directory service: The Contractor will offer a serverless, resilient Active Directory (AD)-compatible service. This is not directory federation. The service will support, at minimum, domain-joining for Windows and Linux compute instances, user identity and group memberships, group policy, and SSO. The service operation will allow the State to maintain compliance with NIST SP 800-53 moderate controls and other applicable federal security standards.	■	

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Vendor Management and Audit			
27.0	Training and certification: The Contractor will offer distinct multilevel training and certification paths for solution architects, operators and developers. The State will have access to free online training for the first-level certification of each path, as well as an introduction to the platform that is suitable for users within the business who need to be acquainted with fundamental platform concepts. Free training may be either first-party or through a Contractor authorized third party. Certifications will require a proctored exam, whether taken in-person or online. Training services will maintain compliance with applicable contractual privacy and security controls.	■	
28.0	Reference architectures: The Contractor will provide publicly published guidance for what constitutes best practices for application architecture on its platform, including guidance for availability, performance and security. In addition, the Contractor will publish a library	■	

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	of reference architectures for at least 20 common application stacks and architectural patterns. The library will also include at least 50 “quick start” provisioning templates, each of which is accompanied by documentation that explains the solution’s implementation and operation.		[REDACTED]
29.0	Marketplace offering: The Contractor will offer a marketplace that contains prebuilt templates and images that are offered by Independent Software Vendor (ISVs) and can be deployed by the State. The Contractor bills the State for marketplace-related consumption as part of the State’s regular cloud bill, and compensates the ISV. The marketplace will be divided into categories, filterable and searchable. Marketplace offerings published by the contractor will follow the terms and conditions set forth in this contract. Any ISV offerings if terms and conditions are not covered under the terms of this contract can still be purchased with approval of a secondary contract with the ISV itself.	■	[REDACTED] ■ [REDACTED] ■ [REDACTED] ■ [REDACTED] ■ [REDACTED]

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31.0	Large-scale provisioning of VMs: The ability to provision a large number of VMs, very rapidly, is critical for disaster recovery, batch computing and scaling on demand. The State also need the assurance that its Contractor has sufficient scale and spare capacity to handle these common use cases. To meet this criterion, the Contractor will support: • Image diversity: Performing disaster recovery, the State will be able to achieve a provisioning time of less than 15 minutes, with an individual VM provisioning failure rate of less than 0.5%, for 1,000 VMs of different sizes, using a diverse set of images. The minimum diversity requirement is 5% Linux images and 5% Windows. Contractor will support at least four VM sizes using the following distribution of minimum sizes (vCPU x approximate GB RAM): 50% 1 x 4 VMs, 25% 2 x 8 VMs, 20% 4 x 16 VMs, and 5% 8 x 32 VMs. • Adequate capacity: The Contractor will have at least one logical data center, in each region, where any customer is able to on-demand provision 1,000	■	

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	VMs. The State will not need to give prior notice or have a capacity reservation. • Reservations: The State will have a means to guarantee that a certain amount of capacity will be available in a particular logical data center, without needing to preprovision that capacity or otherwise pay the fee for a running VM. If a reservation fee is required for this purpose, such fee will be included in pricing schedules, with the minimum term length to be no longer than one year. The maximum size of a reservation will be no less than 1,000 VMs (in aggregate, at least 1,000 vCPUs and 4,000GB of RAM).		
32.0	State-controlled oversubscription: The Contractor will offer at least one VM instance type or tier of service where the State can decide to oversubscribe compute resources. Oversubscribed instances will be priced below standard instances. A standard VM instance cannot be oversubscribed. The Contractor will not sell more virtual compute cores or RAM than physically exists on the host.	■	

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			[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] ■ [REDACTED] [REDACTED]
33.0	Large VM sizes: The State will be able to provision VM with at least 64 vCPUs as well as ~512GB of RAM. This VM configuration will exist within the general-purpose compute service, without the requirement to obtain this VM through a specialized service such as a batch computing or analytics service.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED]

[illegible]

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	infrastructure — compute instances used as container hosts — based on resource needs (such as CPU, memory or availability). - The service will have the capability to be scalable (multinode cluster) and highly available. - The State will be able to customize the OS image that is used on the container hosts (for instance, the Kubernetes worker nodes). The service will support Linux or Windows, but may support only a single OS. - There will be an automatic upgrade capability for the orchestrator, but it is acceptable to require the State to update the compute nodes. - Container hosts will be able to attach volumes from the Contractor's block storage service or file storage service. Container hosts will be able to use the Contractor's virtual networking capabilities, including IP address management, routing, security tools, and firewalls. Private networks will be supported.		[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
35.0	Private image catalog: The Contractor will allow the State to maintain a private image catalog of instances that is not accessible to other cloud customers. The State will be able to provision a compute instance from this catalog. An image can be added to the catalog via customizing and saving the image of an existing compute instance, or importing an image in Open Virtualization Format (OVF or OVA), Virtual Machine Disk (VMDK) or Virtual Hard Disk (VHD) format. The State will be able to use their own custom Linux kernel in the image, and the process and dependencies for doing so will be publicly published.	X	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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36.0	Two-generation OS support: In its catalog of OS images, the Contractor will offer current and “n - 1” long-term support (LTS) versions of the following server operating systems: • At least one enterprise Linux distribution (including Red Hat Enterprise Linux Server) • At least one commonly used free Linux distribution (such as CentOS, Debian or Ubuntu) At least one commonly used container-oriented Linux distribution (such as Red Hat Enterprise Linux Atomic Host)	■	

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				[REDACTED]
37.0	Instance metadata service: The Contractor will offer a local instance metadata key-value store that is API-queryable from within a compute instance. The metadata store contains information about the compute instance, set by the Contractor. At minimum, this will include the instance ID and at least one hostname, location (such as a region or AZ identifier) and network interfaces. The State will also be able to launch a compute instance with arbitrary State-supplied metadata that will be saved in the metadata store.	[REDACTED]		[REDACTED] [REDACTED] ■ [REDACTED] ■ [REDACTED] ■ [REDACTED] ■ [REDACTED] [REDACTED]
38.0	Post-provisioning hooks: The State needs a way to run commands after a newly provisioned compute instance has completed booting. The Contractor will support cloud-init on Linux compute instances, including being able to pass parameters to cloud-init. The Contractor will	[REDACTED]		[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ■ [REDACTED] ■ [REDACTED] ■ [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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	also support an equivalent capability on Windows compute instances.		
39.0	Horizontal autoscaling: Autoscaling is the ability to automatically adjust compute instance capacity, based on specified conditions. This service will have the following capabilities: It will automatically autoscale — provision and deprovision — VMs as necessary. The State will not need to preprovision a compute pool. It will automatically reconfigure the Contractor's load-balancing service when VMs are added or subtracted.	■	
Storage			
40.0	Data migration appliance: The State need a means to cost-effectively, simply and securely migrate data into the Contractor, without encountering network bandwidth or latency challenges. Therefore, the Contractor will offer a service that allows the State to load its data onto a Contractor-owned physical appliance and ship it to the Contractor via traditional express-mailing techniques (e.g., FedEx or United Parcel Service [UPS]). There will be a shipment-receipt endpoint in each of the regions. When the appliance is received, the State will be able to	■	

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	import the data into one or more services that store data. At minimum, the Contractor's object storage service will be supported. The appliance will have the following traits: • Shippable: The appliance will be suitable for shipping, or will be enclosed in a ruggedized case when shipped. Either the appliance or its case will be tamper-resistant or tamper-evident. • Encrypted: All data stored on the appliance will be automatically encrypted. • Large capacity: At least one model of the appliance will have a capacity of 50TB or more. The appliance will have at least one 10 Gbps network interface for the purposes of data transfer. • Eradicated after use: Data on the appliance will be eradicated in accordance with NIST 800-88 media sanitization guidelines, using the purge process, before the appliance is shipped to another customer. Data export appliance: The State need to have a means to cost-effectively, simply and securely retrieve their data from the Contractor, without network constraints. The provider will offer a service by which the State can load cloud data onto a Contractor-owned physical appliance, ship it to the location of their choice via express mail and retrieve the data. The appliance will be shippable, encrypted, large-capacity and eradicated after use. Data transport will comply with all State controls for shipping applicable data.		

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Block Storage Service Criteria			
41.0	Block storage service: The Contractor will offer a scalable and elastic instance-independent block storage service. At minimum, it will have the following capabilities: - A volume will be mountable on any type of compute instance. A nonboot volume will be reassignable to any other compute instance without requiring that either instance be reprovisioned. - The service will support all-solid-state-drive (SSD) storage volume sizes of at least 1TB. The storage will be highly durable by default. All data within the block storage service will be synchronously replicated, so that a minimum of two copies of each volume are stored within a logical data center. The two copies will be stored on different devices. Lower-durability storage may be offered as an option.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
42.0	Flexible block storage volumes: The State will be able to modify attributes of an existing block storage volume, without having to provision a new volume or copy the data. This will include increasing its size and changing its performance tier.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

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43.0	Block storage performance target: The State will be able to choose and achieve an expected I/O performance target for a block storage volume. The Contractor may define targets either in terms of IOPS or throughput. The Contractor will offer either a direct correlation between volume size and an explicit performance target, multiple tiers of block storage with explicit performance targets, or the ability to choose a specific performance target. These targets will be independent of compute instance type, although there may be limits to the options based on instance size and type.	■	
44.0	Encrypted block storage: The block storage service will either encrypt all volumes, or offer the State the option of encrypting specified volumes. An encrypted volume's data is both encrypted at rest, and encrypted in transit between the block storage service and compute instances. Snapshots of encrypted volumes will also be encrypted. All encryption will meet federal FIPS 140-2 standards.	■	
45.0	Block storage snapshots: The block storage service will support crash-consistent snapshots (point-in-time incremental backups) of any storage volume, regardless of whether the volume is attached to a running compute	■	

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	instance. A snapshot can be used to create a new volume with that data.		[REDACTED]
46.0	Block storage data eradication: All eradication methods must be compliance with NIST 800-88 requirements. The block storage service will automatically use one of the following two forms of data eradication: • Immediate eradication: This method immediately overwrites the data associated with a storage volume, when the storage volume is deprovisioned or otherwise released by the State. This allows the State to be assured that the data has been eradicated from the physical disks. The State will be offered a choice to opt into or out of this eradication. • Eventual overwrite: This method guarantees that data deleted by the State cannot be accessed, and that deleted data is eventually overwritten. This means that nobody has access to read a deleted block again until that block is overwritten. Furthermore, if the State deprovisions or otherwise releases a storage volume, those blocks will be overwritten before that storage is made available again (whether to the State or to another customer).	■	[REDACTED]

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Object Storage Service Criteria			
47.0	Object storage service: The Contractor will offer an object storage service that supports an object size of at least 1TB and has the following capabilities: - Objects will be organized into “buckets,” though the object hierarchy may be otherwise flat. The State will be able to assign multiple custom metadata tags to an object. - When an object is stored (PUT), the service will not inform the State the PUT operation has been successful until at least two copies of the object have been successfully written Both copies may be in the same data center. In regions with AZs, the State will be able to choose a tier of object storage where the object is stored in at least two AZs within the region. In regions without AZs, the State will be able to choose a tier of object storage where the object is stored in at least two logical data centers in the same country. - The State will be able to provide a hash-based integrity check (using a common algorithm such as MD5) for the object in the header of a PUT request. The service will compute this on the uploaded object, and if the hashes do not match, it will discard the object and return an error. - When an object is retrieved (GET), the State will be able to do so via	■	

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	HTTPS. The service will send, in a header, an MD5 hash (or equivalent) for the object, so the State can verify the object's integrity.		
48.0	Object permissions: The object storage service will support an RBAC scheme for object or bucket access, including GET, PUT and LIST permission. These RBAC policies may be applied at either the bucket or individual object level. The policies will support granting access to an individual user, granting access to groups of users (based on an attribute such as a role) and granting access to users in a third-party organization. This RBAC will be integrated with the IAM service used by the State.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
49.0	Durable object storage: To achieve greater durability, the State will be able to choose a tier of object storage that automatically replicates the object to three or more locations, in at least two logical data centers. Data integrity will be periodically verified through the use of checksums or the like. Furthermore, the Contractor will implement erasure coding or a similar scheme that tolerates multiple concurrent failures. Erasure coding is a method of data protection by which data is fragmented, expanded and encoded with redundant data pieces, and then stored across a set of different locations.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
50.0	Encrypted object storage: The object storage service will either offer service-side encryption (SSE) of all objects at rest, or allow the State to opt into SSE of specific objects (or buckets of objects) at rest. The SSE capability will support BYOK, and may support HYOK.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED]

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			[REDACTED]
51.0	Cross-region object replication: The State will be able to select a tier or opt-in capability of object storage that automatically replicates an object into another region of their choice. Such replication will be near-real-time but may be asynchronous. If the Contractor offers encrypted object storage, this capability will be able to replicate encrypted objects.	■	[REDACTED]
52.0	Resilient object transmission: The object storage service will provide a capability for resumable object uploads, so that a network failure does not require reuploading the entire object. This capability may be API-only. For example, this may be accomplished via an API	■	[REDACTED]

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	that allows uploading multiple parts of an object and reassembling them into a single object. Furthermore, the API will support downloading a specified portion of an object.		
53.0	Log object operations: The Contractor will enable the State to log object storage access requests; this logging may be best-effort, without guarantees for timeliness or completeness of logs. In addition, the Contractor will enable the State to log and audit operations on objects. This will include, at minimum, any operation that creates, deletes, or alters an object or bucket. Audit and access operations may be separate logs.	■	
File Storage Service Criteria			
54.0	File storage service: The Contractor will offer a file storage service that allows the State to obtain network file shares. The service will have the following capabilities:	■	

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	- A file system within the service will be mountable on any type of compute instance, including compute instances in a different region. - The service will support either NFSv3 or SMB 3.0 (or higher). The service may also support older NFS or SMB versions. For NFSv3, the Contractor will support a means of restricting what hosts can access the file share. For SMB 3.0 or higher, the Contractor will support built-in authentication, ACLs or directory service integration. - The storage will be highly durable by default. All data within the file storage service will be synchronously replicated, so that a minimum of two copies of each file are stored within a logical data center. The two copies will be stored on different devices. Lower-durability storage may be offered as an option. - The storage and communications will meet federal encryption requirements.		
Network			
55.0	Virtual networking: The Contractor will offer software-defined LAN and WAN services, allowing the State full flexibility to configure a multi-segment hierarchical network topology that is independent of compute instances. At minimum, the following capabilities will be supported: Routing: The State will be able to define a gateway and custom routing for each virtual network. The State will be able to have at least 20 virtual networks per region.	■	

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	Subnets: The State will be able to define subnets within a virtual network. The State will be able to have a minimum of 25 subnets per virtual network. Private networks: The State will be able to define virtual networks that are fully isolated and not externally routable, with their choice of RFC 1918 private IP address blocks. Traffic routed between the State's private networks within the same logical data center will have the capability to remain within private IP space. The traffic will not cross the public internet.		
56.0	Private networks between regions: All traffic between the Contractor's general-purpose data centers will remain on the Contractor's private WAN (though this does not require private fiber). It cannot use the internet, even if the traffic is tunneled. Furthermore, traffic routed between any of the State's private networks, including those in different logical data centers or different regions, will have the capability to remain within private IP space.	■	
57.0	Private networks for compute instances: The Contractor will support the following capabilities, to the extent that the OS of the compute instances permit: - All compute instances will be able to have only private IP addresses. - All compute instances will be able to have at least two IP addresses, and simultaneously have public and private IPs. The State will be able to independently route each IP.	■	

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	All VM instances will be able to have multiple virtual network interface cards (vNICs), each with its own Media Access Control (MAC) address.		
58.0	Private networks for data-handling services: The State will be able to keep traffic between their compute instances and data services within a private network. The following services will support private networks, if the Contractor has such a service: the key management service, the object storage service, at least one relational DBaaS, and the data warehouse service.	■	
59.0	Private networks between organizations: The State will be able to route traffic between two private networks within two different partitions. The State will also be able to route traffic between two private networks that belong to different cooperating organizations (including accounts held by completely separate entities). Traffic routed between private virtual networks will have the capability to remain within private IP space.	■	

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60.0	Private State connectivity: The State will be able to connect their private WAN with their cloud virtual networks, as follows: The State will be able to obtain private WAN connectivity into the cloud Contractor via a carrier or colocation facility. (WAN connectivity can take a number of forms, such as Ethernet, Multiprotocol Label Switching [MPLS], virtual private LAN service [VPLS] or a direct cross-connect.) Contractors will offer at least two connectivity facility options per general-purpose region. The State will be able to have at least two private connections per region, and per virtual network. Service capabilities, such as routing features, will support both multisite customers with multiple regional gateways, as well as customers that want redundant multicarrier connectivity into a single logical data center.	[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] - I [REDACTED] - I [REDACTED] - I [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
61.0	Site-to-site VPN gateway: The Contractor will offer a site-to-site VPN service that connects the State's virtual networks to their premises over the public internet, via an IPsec or SSL/TLS VPN tunnel. A virtual network will be able to have at least two VPN connections. VPN connections will support redundant Contractor termination points.	[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
62.0	Transit hub: The State will be able to route a virtual network's traffic through a centralized private network connection point for use with centralized gateways or security appliances that require a hub-and-spoke topology. The Contractor will offer a transit routing service capability that allows the State to redirect virtual network traffic through a security function, such as a	[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

[illegible]

[illegible]

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			[REDACTED]
65.0	Layer 7 load balancing: The Contractor will offer a Layer 7 LB service that distributes HTTP and HTTPS traffic to compute instances and has the following capabilities: • Basic load balancing: The LB supports public and private networks, basic load distribution, session affinity and SSL offload. (See the "Web load balancing" criterion.) • URI-based routing: The LB can route requests based on State-defined unified resource identifier (URI) patterns. • X-Forwarded-For: The load balancer will support the configuration and use of X-Forwarded-For by the State if needed. HTTP/2: The LB can deliver content via HTTP/2 to clients that support HTTP/2, while supporting a HTTP/1.1 back end.	■	[REDACTED]

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			[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
66.0	Static IP addresses: If a compute instance is dynamically assigned an IP address, that address will remain the same across the lifetime of the instance (unless the State wants to change it). Furthermore, the State will be able to obtain an indefinitely persistent private or public IP address that can be assigned to a compute instance or LB, and that is reassignable on demand. The Contractor must be able to support portable virtual NICs, where the State can opt to move a vNIC from one instance to another, which reassigns the MAC address and associated IP addresses.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
67.0	DNS service: The Contractor will offer DNS as a service. The State will have the option of designating this DNS service as an authoritative (i.e., primary or secondary) DNS for a zone. Furthermore, if the State uses the Contractor's IP addresses, the State will be able to create reverse DNS (PTR) records for those IP addresses when they are assigned to compute instances.	■	[REDACTED] [REDACTED] [REDACTED] [REDACTED]

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			■ [REDACTED] [REDACTED] ■ [REDACTED] [REDACTED] ■ [REDACTED]
Software Infrastructure Services			
68.0	Relational DBaaS: The Contractor will offer a serverless relational database as a service (rDBaaS). Minor version updates and major version upgrades may have different levels of automation and necessary downtime. The State will be able to configure and schedule automated backups with a retention policy, as well as export a database dump. At a minimum, the service will support one common open-source database and one common proprietary database.	■	[REDACTED]
69.0	Relational DBaaS read replicas: At least one of the Contractor's serverless rDBaaS offerings will scale beyond a single instance. The Contractor will either provide automatically scaled capacity or the State will be able to opt into database read replicas.	■	[REDACTED]
70.0	Relational DBaaS synchronous replication: To achieve higher availability, at least one of the Contractor's serverless rDBaaS offerings will offer synchronous replication of a State's database into another logical data center. This may be an active-active or automatic failover architecture, but the application will see only a single endpoint.	■	[REDACTED]
71.0	Data warehouse service: The Contractor will offer a serverless data warehouse service. It will be ANSI SQL-compatible, and there will be ODBC and JDBC drivers. There will be documented integrations with common	■	[REDACTED]

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	business intelligence (BI) tools, such as MicroStrategy and Tableau Software.		
72.0	Functions as a service: The Contractor will offer a service for the execution of event-triggered functions. This seamlessly scalable and resilient serverless offering executes State-provided code on demand with support for at least three programming languages. It charges only for the resources used, has a limited execution time, scales rapidly, handles concurrency and availability across networks, and provides an error-handling mechanism. The Contractor's monitoring or notification service (i.e., a service that invokes or handles alert-generation mechanisms) will be able to invoke a function, so that functions can be used for operational needs.	■	██████████

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